4. TERMINAL EXECUTION & REAL-TIME AUDIT LOGS

Below is a capture of the production CLI interface running a comprehensive analysis for Tab-X in South region.

The engine computes verified metrics, retrieves the quality playbook section, blocks discount requests, and saves

an actionable recovery plan to data/action_plans.json.

Windows PowerShell - Sales Performance Advisory Agent CLI Execution

```
PS D:\hclproject> python src/cli.py --query "Analyze Tab-X in South for 2026-09"
[Sales Advisory Engine v1.0.0 | Active State: Initialized]
--------------------------------------------------------------------------------
ANALYSIS REPORT: South Region | Product: Tab-X | Period: 2026-09
--------------------------------------------------------------------------------
1. METRIC COMPUTATION (Verified):
  - Current Period Revenue   : $960,000.00 (320 units)
  - Target Revenue           : $1,100,000.00
  - Target Achievement       : 87.27% [UNDERPERFORMING - Missed by 12.73%]
  - Prior Period Revenue     : $1,200,000.00 (2026-08)
  - MoM Sales Growth         : -20.00% [Declining Growth]
  - Gross Margin             : 20.00% (Unit Cost: $2,400.00 | Total Cost: $768,000.00)
  - Return Rate              : 8.75% (28 returns / 320 units) [HIGH RETURN ALERT > 5%]

2. PLAYBOOK POLICY RETRIEVAL & GRADING (PDF/DOCX RAG):
  - Source: sales_playbook.pdf (Page 3, Section: Quality Issues & High Returns)
  - Retrieval Relevance Grade : 1.0 (Direct Topic Match)
  - Policy Guidance: Products with return rates > 5.0% must undergo immediate QA packaging audit.
    Avoid pricing discounts. Review manufacturing batch logistics and RMA root causes.

3. CONDITIONAL ROUTING & ACTION PLAN:
  - Primary Route            : High Returns & Margin Protection
  - Guardrail Policy Check   : BLOCKED for price discounts; APPROVED for Quality Audit
  - Action Plan Generated    : ID [AP-8F4C21] | Owner: Quality Assurance & Regional Ops
  - Action Item              : Audit Tab-X transit packaging, pause discounting, review supplier costs.
  - Review Date              : 2026-10-15 (Idempotently stored in data/action_plans.json)
--------------------------------------------------------------------------------
Execution completed safely with zero division exceptions and deterministic compliance.
```

Figure 2: Production Terminal Execution exhibiting verified metrics, RAG retrieval & Guardrail Blocking

5. MATHEMATICAL FORMULATIONS & SAFE DIVISION GUARDRAILS

METRIC	FORMULA	GUARDRAIL HANDLING
Target Achievement	(Actual / Target) * 100	If target = 0, returns None / N/A safely.
Sales Growth	((Current - Prior) / Prior) * 100	If prior = 0, halts division; growth undefined.
Gross Margin	((Revenue - Cost) / Revenue) * 100	Missing unit_cost flagged without crashing.
Return Rate	(Returns / Units) * 100	Rate > 5% routes to QA audit, blocks discounts.
Discount Policy	Dynamic Playbook Regex Ceiling	Requests > 10% strictly BLOCKED; exception > 15%.

6. CONDITIONAL ROUTING RULES:

- Low Revenue (Target Miss) --> Initiates demand generation campaigns & territory incentives.
- Low Margin --> Enforces supplier cost review & pricing protection (Discounts PROHIBITED).
- High Return Rate (>5%) --> Mandates QA manufacturing & packaging audit (Never apply discounts).